

Eren Balkış

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Education

Istanbul University, Computer Science (B.Sc.)

September 2023 – June 2027

- GPA: 3.27/4.0
- **Key Courses:** Artificial Intelligence, Machine Learning, Object-Oriented Programming, Computer Networks

Experience

IT Intern, Galata Wind – Istanbul, Turkey

August 2026 – Present

- Developed a custom-built network monitoring system from scratch using SNMPv2 and ICMP to track 86 servers/network devices and 2 public websites. Achieved a scanning time of approximately 28 seconds for all 88 assets utilizing a semaphore-limited concurrent polling architecture.
- Implemented an email alerting system with a flap damping mechanism triggering upon 3 consecutive failures, alongside an automated 24-hour summary report sent daily at 09:00. Configured the poller as a systemd service and developed a secure import infrastructure managing the device inventory via CSV as the single source of truth.

Projects

CardioRisk AI: Heart Disease Risk Prediction

Github – Live App

- Built a machine learning pipeline featuring outlier removal via **IQR**, **one-hot encoding**, and normalization using **StandardScaler** on 303 patient records and 13 clinical features. The **Logistic Regression** model achieved approximately **90%** accuracy on the test set.
- Converted the model into a **FastAPI** (REST) service validated with **Pydantic** and deployed it on **Render**. Ensured consistency by utilizing the same preprocessing pipeline for both training and inference. Addressed the reverse labeling present in the dataset correctly within the API.
- Developed a step-by-step user-guided **Next.js** interface (featuring an animated radial risk indicator, a results panel, and an advanced mode with 7 additional parameters) and deployed the end-to-end application on **Vercel**.
- **Technologies Used:** Python, Scikit-learn, Pandas, FastAPI, Pydantic, Next.js, Tailwind CSS, Render, Vercel

Smart Transit AI: Crowd Analysis for Public Transit Stops

Github - Live App

- Developed a **CSRNet**-based **density map** regression system to detect overcrowding at transit stops. Trained the model on the **ShanghaiTech** Part B dataset, enabling accurate crowd counting in dense and overlapping scenarios where object detection methods like **YOLO** typically fail.
- Built a decision mechanism that calculates the headcount from the sum of the density map and triggers an extra bus dispatch alert when a threshold is exceeded (**over 50 people**). Developed a side-by-side interface displaying the original image and heat map using **Streamlit** and deployed it on **Hugging Face Spaces**.
- **Technologies Used:** Python, PyTorch, CSRNet, VGG-16, Streamlit, Computer Vision, Hugging Face Spaces

Technical Skills

Tools and Technologies: Python, C++, SQL, Scikit-learn, Pandas, NumPy, Matplotlib, SNMP, ICMP, TCP/IP, Linux (Ubuntu, Zorin), Git/GitHub, FastAPI, Streamlit, Vercel, Render, MS Office (Word, Excel, PowerPoint)

AI and Methodologies: LLM, Supervised Learning, Predictive Modeling, EDA, Hyperparameter Tuning

Certifications

DataCamp: Intermediate SQL | **Kaggle:** Intermediate Machine Learning **Cisco:** Networking Basics |
Turkcell Geleceği Yazanlar: Python 401 | **Claude Academy:** Claude Code 101